

Raunak Shukla

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AI Engineer with hands-on experience building scalable AI systems and deploying LLM driven applications to production.
Proven track record across full-stack development, cloud infrastructure, multi-agent architectures & RAG pipelines.

EDUCATION

New York University - Tandon School of Engineering

Master of Science (MS) in Computer Science

May 2025

GPA: 3.73/4.0

Relevant Coursework: Algorithms, Software Engineering, Machine Learning, Data Science, Data Visualization, Big Data, Network Security

Vellore Institute of Technology, Vellore

Bachelors of Technology (B.Tech) - Computer Science and Engineering

May 2023

GPA: 8.44/10

WORK EXPERIENCE

AI Engineer - BDR (OpenAI, RAG, AWS, Python, React Native)

June 2025 - Present

- Engineered a **retrieval-augmented generation (RAG)** pipeline using **OpenAI embeddings** and **Pinecone** vector database to index and retrieve inspection and thermal analysis data, enabling context-aware inputs for downstream analysis.
- Deployed a dockerized ML pipeline on **AWS Fargate** to process large-scale thermal and drone telemetry data for anomaly detection and cost modeling, reducing runtime by **~40%** through parallelization and resource optimization.
- Designed and implemented an LLM-based report generation pipeline to convert structured inspection and telemetry data into standardized natural language reports, enabling automated and consistent output generation.
- Optimized LLM performance through prompting strategies (few-shot, schema-constrained outputs) and implemented output validation and structured parsing to improve response consistency, reliability, and downstream system integration.
- Architected a cross-platform (Android & iOS) **mobile application** to capture site data and generate AI-powered inspection reports using the LLM pipeline with real-time data synchronization via **GraphQL** subscriptions, scalable to **1,000+ users**.

Software Engineer Intern - BDR (Next.js, TypeScript, PostgreSQL, Supabase, Stripe)

March 2025 - June 2025

- Built and launched the company's [website](#) from scratch, implementing server-side rendering, dynamic routing, and optimized content delivery, improving performance and page load times by **~30%**.
- Designed a **relational database schema** to store user accounts and inspection reports, enabling secure, account-specific access through authenticated workflows, and integrated a **payment system** to support end-to-end transaction flows.

Machine Learning Intern - Feynn Labs (Python, Scikit-learn, XGBoost, Tableau)

Oct 2022 - Dec 2022

- Developed and evaluated machine learning models for market segmentation using **clustering techniques** and **gradient boosting**, achieving **~92%** accuracy in predicting customer segments.
- Conducted exploratory data analysis and modeled data pipelines, leveraging **Tableau** for visualization to generate actionable insights, optimizing marketing strategy and improving data-driven decision making.

PROJECTS

Multi-Agent Marketplace (Facebook Marketplace-style) (OpenAI, FastAPI, PostgreSQL, Redis, React, Vite)

- Engineered a multi-agent marketplace where autonomous buyer/seller agents search listings, exchange messages, and negotiate prices using LLM-driven decision making, powered by **semantic search** with **OpenAI embeddings** for context-aware retrieval.
- Created a real-time copilot chat experience using **WebSockets** with **persistent session memory**, enabling users to browse listings, refine queries, and interact with agents through end-to-end negotiation workflows.

TaskPilot: Agentic Workflow Engine (LangGraph, Multi-Agent Systems, FastAPI, Redis, OpenAI)

- Streamlined a multi-agent orchestration system enabling autonomous agents to plan, execute, and validate tasks with structured reasoning, contextual memory, and task coordination.
- Developed a real-time execution framework with streaming outputs and persistent state, supporting error handling and recovery while simulating end-to-end DevOps workflows through agent-driven automation.

TECHNICAL SKILLS

Languages: Python, SQL, Javascript, TypeScript, Java, C++

AI/ML: TensorFlow, PyTorch, MCP, Feature Selection, Predictive Modelling, Scikit-Learn, Pandas, Numpy

GenAI & LLMs: OpenAI API, RAG, Multi-Agent Systems, LangGraph, Prompt Engineering, LLM Fine-Tuning (SFT, PEFT)

Backend & Systems: FastAPI, Node.js, Express, REST APIs, GraphQL, Redis, Distributed Systems, Django, Spark, Flask

Frontend: React.js, Next.js, React Native, Tailwind CSS | **Software Dev:** Agile, Scrum, Code Reviews, Unit Testing

Cloud: AWS, Azure, GCP, Docker, Kubernetes, CI/CD, Jenkins | **Databases:** PostgreSQL, MongoDB, Vector DB (Pinecone)

Tools: VS Code, Cursor AI, Open AI, Claude, Android Studio, XCode, Kafka, Terraform, Airflow, Snowflake, Github, Gitlab